

Fold Six Napkins Decoratively SOP

In one episode you fold six napkins and put them on three plates. Napkins come off the supply stack one at a time. Each one is folded on the fold spot, then put on a plate. Each plate ends up with one triangle lying flat and one fan leaning on it. At the end the supply spot is empty and the fold spot is clear.

The triangle and the fan are two different folds. They share nothing. Both start from the open napkin lying square on the fold spot.

- A **triangle** is one corner folded onto the corner across from it, then creased.
- A **fan** is pleats folded back and forth across the open napkin, then folded in half at the middle.

A napkin gets one fold or the other. Never both.

Most of the time the left gripper holds the cloth and the right gripper does the work. Four jobs break that rule, and only these four:

- Both grippers lift a napkin off the supply stack.
- Both grippers swing each pleat, one on each end.
- Both grippers carry a triangle napkin to its plate and lay it down.
- The left gripper fills the left plate with a fan.

Never turn a napkin in the air. Never drag it across the fold spot. The only thing that drags is a crease. A gripper only opens over the fold spot or over a plate.

Setup

Do both checklists before you start an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the three plates along the far side, the fold spot in the middle near the front edge, and the supply stack on the left.
3. Both arms are at home with grippers open.
4. The tabletop is clear except for the areas below.
5. Nothing blocks the paths between the supply stack, the fold spot, and the three plates.

Materials checklist

1. Six matching square cloth napkins. Pressed flat, dry, no stains, no tears.
2. The six napkins sit in one neat stack on the supply spot. Each lies square, with one edge facing the front edge. No napkin hangs over the one below it.
3. The napkins pull apart easily. Replace any napkin that sticks to the one under it.
4. The fold spot is clear: the middle of the tabletop, near the front edge, is clean, dry, and has nothing on it. Cloth does not slide on it when a gripper presses and drags.
5. Three matching plates sit in a row along the far side. They are spaced apart so no plate touches another. All three are empty, right side up, and big enough that a folded napkin does not hang over the rim.

Workspace layout

- **Supply spot:** the stack of six napkins on the left of the tabletop, in line with the fold spot. Napkins are picked up here and nowhere else.
- **Fold spot:** the middle of the tabletop, near the front edge. The napkin is folded on the tabletop itself. Both arms reach it. Every fold and every crease happens here. It holds one napkin at a time. It is clear at the start and at the end.
- **Left plate, middle plate, right plate:** three plates in a row along the far side, in the middle. They are named left to right and filled left to right. Nothing else is ever put on them.

The three areas sit in a line from the far side inward. The plates are furthest away. The fold spot is in front of them, in the middle of the tabletop near the front edge. The supply stack is off to the left, in line with the fold spot.

Vocabulary

- **Arm jobs:** the left gripper holds the cloth. The right gripper folds, creases, and fills the middle and right plates with a fan. Four jobs work differently: lifting a napkin off the supply stack, swinging a pleat, carrying a triangle to its plate, and filling the left plate with a fan.
- **Turn:** which napkin you are on, 1 to 6. Turn 1 is the first napkin off the stack. The turn decides the fold and the plate.
- **Square:** the napkin lies flat and open on the fold spot with one edge facing the front edge. Every napkin lands this way, no matter which fold it gets.
- **Napkin edges:** near, far, left, and right. Named after the side of the tabletop they face.
- **Napkin corners:** near left, near right, far left, and far right.
- **Square corner:** on a folded triangle, the corner where the two short sides meet. It sits at the far left.
- **Hold:** a closed gripper pressed straight down on the cloth to keep it still. A hold does not pinch the cloth and does not move.

- **Fold:** a gripper pinches a corner or an edge. It lifts it off the cloth below, carries it over, lowers it, and lets go.
- **Crease:** the closed right gripper presses down on a new fold edge. It drags along the edge in one stroke, end to end, without lifting off.
- **Fold edge:** the folded side. It is closed. The other sides are open.
- **Triangle napkin:** one corner folded onto the corner across from it, creased, and nothing more. Napkins 1, 3, and 5.
- **Fan napkin:** creased pleats across the open napkin, then the gather. It stays folded shut on the fold spot. It opens into a fan when it is leaned on the triangle. Napkins 2, 4, and 6.
- **Working edge:** the near edge of the open napkin. Both grippers hold this same edge, one on each end, and swing it back and forth for every pleat.
- **Pleat:** one swing of the working edge. Both grippers pinch it, lift it, carry it the other way from the last swing, and lay it down. That makes a new crease.
- **Swing:** how far a pleat travels, away from the front edge or back toward it. Every swing goes about the same distance, so the bands come out even.
- **Band:** the strip of cloth between one crease and the next. The pleats leave bands that look about the same width.
- **Zigzag:** the shape the pleats make when you look at the napkin from the side. The cloth goes up, back, up, back.
- **Gather:** the last fold on a fan napkin. The pleated band is folded in half at the middle. The pleats sit inside each other and the napkin stays shut. It opens later, when it is leaned.
- **Gather point:** the folded middle that the gather makes. All the pleats are held together there. It is the only place a fan napkin is picked up.
- **U-lift:** how a triangle napkin leaves the fold spot. Both grippers pinch the fold edge, one on each end, and lift together. The cloth hangs in a U between them and the layers stay together. It is the only way a triangle is picked up.
- **Plate gripper:** the gripper that fills a plate with a fan napkin. Left plate, left gripper. Middle and right plates, right gripper. That same gripper picks the fan up off the fold spot and carries it over. A triangle napkin does not use a plate gripper. Both arms carry it, whichever plate it is going to.
- **Setting:** one finished plate. A triangle lying flat with a fan leaning on it. The episode ends with three settings.
- **Resting:** the napkin is down and takes its own weight. It stays put for 2 seconds after the gripper lifts off, and no layer pops open.
- **Release point:** the fold spot and the three plates. A gripper opens nowhere else.

Steps

One napkin runs one step from start to finish: off the supply stack, onto the fold spot, folded there, then onto its plate. Only then does the next napkin come off the stack. A napkin runs **either** Step 1 **or** Step 2, never both. Napkin 1, 3, or 5 runs Step 1, the triangle. Napkin 2, 4, or 6 runs Step 2, the fan.

The turn says which plate. Plates are filled left to right, two napkins each. Turns 1 and 2 go to the left plate. Turns 3 and 4 go to the middle plate. Turns 5 and 6 go to the right plate. The odd turn brings the triangle. The even turn after it brings the fan that leans on that triangle.

The two steps alternate all the way through the supply. Step 1 on napkin 1, Step 2 on napkin 2, Step 1 on napkin 3, and on to napkin 6. The order never changes. Always take the top napkin off the stack, and finish it, plate and all, before the next one comes off.

Step 1: Fold a triangle napkin and lay it on its plate

Only for a **triangle napkin**, which is napkin 1, 3, or 5. A fan napkin runs Step 2 instead.

Goal: a creased triangle lying flat on its plate, the same way every time.

1.1 Take the top napkin

- The **left gripper** pinches the top napkin's **far left corner**. The **right gripper** pinches its **near right corner**. Take the top napkin only.
- The two grippers hold opposite corners, so the napkin hangs flat between them.
- Both grippers lift it straight up, clear of the stack.

1.2 Lay it out on the fold spot

- Both grippers carry it level to the right and lower it flat onto the fold spot.
- Lay it square, with one edge facing the front edge.
- Let go once the napkin is resting. Lift both grippers clear.

Check: the napkin is flat and open. One edge faces the front edge, not a corner. No corner is folded under. Nothing hangs off the fold spot.

If it lands crumpled or turned, lift it by the same two corners and lay it down again. If a second napkin comes up with it, put both on the fold spot, take the extra one back to the supply stack, and lay the other one out again.

1.3 Hold the napkin

- The **left gripper**, closed, presses down on the **far left corner**. It stays there through the fold and the crease.

1.4 Fold the near right corner onto the far left corner

- The **right gripper** pinches the **near right corner** and lifts it off the cloth.

- It carries the corner across the napkin and lowers it onto the **far left corner**, where the left gripper is holding. Let go once it is resting.

Lift the corner off the cloth before you carry it. Drag a corner across the fold spot and it pulls the whole napkin with it, so the fold lands crooked.

1.5 Crease the fold edge

- The new fold edge runs from the **near left corner** to the **far right corner**.
- The **right gripper**, closed, presses down on that edge at the **near left corner**. It drags along the edge left to right, in one stroke, all the way to the **far right corner**.
- Both grippers lift clear.

The crease always runs left to right. The left gripper is holding the far left corner, which is off the fold edge, so it never blocks the stroke.

Check: the napkin is a triangle. The moved corner sits on the far left corner, not off to one side. The long fold edge runs from the near left to the far right. The whole edge is creased flat. No layer has popped open.

If the corners are clearly off, open the napkin back out square and fold it again.

A triangle travels on both arms, from the fold spot all the way down onto the plate. It does not use a plate gripper.

1.6 U-lift the triangle off the fold spot

- The **left gripper** pinches the **near left end** of the **fold edge**. The **right gripper** pinches the **far right end** of that same edge. That is the long folded side, where the crease is.
- The two grippers hold opposite ends, so the cloth hangs in a U between them and the layers stay together. Both grippers lift the triangle level off the fold spot, as a pair. Never pick a triangle up on one arm.

1.7 Lay the triangle on its plate

The plate is still empty. Turn 1 goes to the left plate, turn 3 to the middle plate, turn 5 to the right plate.

- Both grippers carry the triangle level to its plate, still holding the two ends of the fold edge. They move together, so the U keeps its shape and neither end runs ahead.
- They lower the triangle flat onto the middle of the plate. The long fold edge runs from the near left to the far right. The square corner sits at the far left. Both let go together, once it is resting.
- Lift both grippers clear. Every triangle lands the same way.

Carry the napkin level. Tip it up on an open edge and the layers slide apart on the way.

Check: the triangle is on the plate the turn says. It lies flat and centered, the same way as the ones on the other plates. Nothing hangs over the rim. No layer has popped open. The fold spot is clear again.

If it lands crooked or in the wrong place, lift it clear and lay it down again. If a layer popped open, take it back to the fold spot, open it out square, fold it again, and bring it back.

Step 2: Fold a fan napkin and lean it on the triangle

Only for a **fan napkin**, which is napkin 2, 4, or 6. A triangle napkin runs Step 1 instead. A fan napkin never gets the triangle fold.

Goal: a pleated fan leaning on the triangle already on its plate, spread open with nothing holding it up but the triangle.

2.1 Take the top napkin

- The **left gripper** pinches the top napkin's **far left corner**. The **right gripper** pinches its **near right corner**. Take the top napkin only.
- The two grippers hold opposite corners, so the napkin hangs flat between them.
- Both grippers lift it straight up, clear of the stack.

2.2 Lay it out on the fold spot

- Both grippers carry it level to the right and lower it flat onto the fold spot.
- Lay it square, with one edge facing the front edge.
- Let go once the napkin is resting. Lift both grippers clear.

Check: the napkin is flat and open. One edge faces the front edge, not a corner. No corner is folded under. Nothing hangs off the fold spot.

If it lands crumpled or turned, lift it by the same two corners and lay it down again. If a second napkin comes up with it, put both on the fold spot, take the extra one back to the supply stack, and lay the other one out again.

The pleats start here, on the open napkin.

Both grippers swing each pleat. The **left gripper** takes the left end of the working edge. The **right gripper** takes the right end. They lift, swing, and lower as a pair, so the edge stays straight.

Each swing goes the other way from the one before. Away from the front edge, back, away, back, on and on to the far edge. Every swing travels about the same distance.

The crease works the normal way. The left gripper holds one end down. The right gripper drags the crease.

2.3 Swing the first pleat and crease it

- Both grippers pinch the **working edge**, one on each end, and lift it off the cloth.
- Both carry it away from the front edge and lower it partway up the napkin. Let go once it is resting.
- The **left gripper**, closed, presses down on the left end of the new crease and holds.
- The **right gripper**, closed, presses on the crease and drags right, all the way to the right end.

2.4 Keep swinging pleats to the far edge

Repeat the same pleat, the same way, until the napkin is used up.

- Both grippers pinch the **working edge** again and lift it off the cloth.
- Both swing it the other way from the last swing, about the same distance, and lower it. Let go once it is resting.
- Crease it the same way. Left gripper holds the left end, right gripper drags to the right end.
- Swing again, the other way, and crease again. Keep going until the working edge reaches the far edge and no flat cloth is left to swing.
- Both grippers lift clear.

Every pleat goes the other way from the last one. Two swings in a row the same way roll the napkin up flat, and the zigzag is gone.

Keep the swings the same length. A short swing after a long one leaves one band much thinner than the rest, and the fan looks uneven. Do not stop with flat cloth still lying past the last crease, and do not stretch the last swing to swallow what is left.

Both ends of the edge move together. If one gripper gets ahead of the other, the crease lands at an angle.

Fix a bad pleat before the gather. Lift the edge back up, swing it again, and crease it. The gather folds the pleats onto each other, and after that you cannot reach a bad pleat without opening the whole napkin back out.

2.5 Gather the band into a fan

The napkin is now a wide band running left to right. Fold it in half at the middle and the pleats sit inside each other. The napkin stays shut here. It does not open on the fold spot, and it is not supposed to. The gather is what lets it open in Step 2.7, when it leans on the triangle.

- The **left gripper**, closed, presses down on the **middle** of the band and holds. That spot is the fold line. It does not move.
- The **right gripper** pinches the **right end** of the band, lifts it off the fold spot, and carries it left over the top of the left gripper.

- It lowers the right end onto the **left end** of the band. The two halves sit on each other and the pleats tuck together. Let go once it is resting.
- Both grippers lift clear. The napkin stays folded shut, the two halves resting on each other.

The left gripper holds the middle and nowhere else. Hold it off to one side and the fold line is off the middle. Then one half of the fan is longer than the other and the pleats do not tuck together.

Do not crease this fold. Press it flat and the pleats lock together, and the napkin will not open when it is leaned on the triangle.

Check: the napkin is folded shut with the two halves on each other, about the same length. The pleats sit inside each other, not crossed. The fold is at the middle, and it is soft, not creased.

If the halves are clearly uneven, or the gather fold got creased flat, open the fold back out, straighten the band, and fold it in half again.

A fan travels on one arm, the **plate gripper**: the **left gripper** fills the left plate, the **right gripper** fills the middle and right plates, so neither arm has to reach across the table for a fan.

2.6 Pick the fan up by the gather point

- The plate gripper pinches the **gather point**, the folded middle, and lifts it level off the fold spot. Never lift a fan by a pleat or by an open end.

2.7 Lean the fan on the triangle

This plate already holds the triangle from the turn before. Turn 2 goes to the left plate, turn 4 to the middle plate, turn 6 to the right plate.

- The plate gripper carries the fan level to its plate.
- It sets the **gather point** down on the near half of the triangle. Then it lets the open side lean back onto the triangle. The two halves come apart as it leans, and the pleats spread into a fan. This is where the napkin opens. It does not flatten out.
- Let go once the fan is resting and leaning. Lift the gripper clear.
- Nothing presses the fan down. The triangle holds it up on its own.

Carry a fan level by the gather point and the pleats stay tucked together the whole way.

Check: the fan is on the plate the turn says, leaning on that plate's triangle with its pleats standing and spread. Nothing hangs over the rim. Neither napkin has slid off the plate. The fold spot is clear again.

If the fan lands crooked or in the wrong place, lift it clear and lean it again. If it will not open when it leans, take it back to the fold spot, open it out square, fold it again, and bring it back.

Step 3: End the episode

Goal: all six napkins folded and on their plates, and nothing left behind.

- Check that all six napkins are folded and on their plates. Each of the three plates holds one triangle lying flat with one fan leaning on it. Six napkins, three settings, none missing.
- Check that the supply spot is empty and the fold spot is clear. A napkin left on either one means a turn was never finished.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the six napkins off the three plates and open each one out flat.
2. Press every napkin flat again so no fold line from the last episode is left in the cloth. Do not restack a napkin that still shows an old crease.
3. Restack the six pressed napkins neatly on the supply spot, each lying square with one edge toward the front edge. Change the order the napkins go back in each time.
4. Replace any napkin that is torn, stained, damp, or sticking to its neighbor.
5. Wipe all three plates, dry them, and put them back empty in a row along the far side.
6. Leave the fold spot clear, in the middle of the tabletop near the front edge.
7. Wipe the tabletop.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for training. Do not use it as a label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong fold for the napkin's turn

- **Visible cue:** a napkin on turn 1, 3, or 5 is pleated into a fan. Or a napkin on turn 2, 4, or 6 is folded into a triangle. Or one napkin gets both folds: folded corner to corner, then pleated.
- **SOP rule broken:** Steps 1 and 2. The two steps alternate and the order never changes, and a napkin runs Step 1 or Step 2, never both.
- **Coaching note:** say the fold out loud before you take the napkin. Odd turns are triangles.

Violation: Wrong napkin taken from the supply

- **Visible cue:** a napkin is pulled out from under another, slid out sideways, or taken from the middle of the stack. Or two come off together and both go onto the fold spot or a plate. Or a second napkin is picked up while one is still on the fold spot.
- **SOP rule broken:** Steps 1.1 and 2.1. Take the top napkin only. The fold spot holds one napkin at a time, and a napkin goes on its plate before the next one is picked up.
- **Coaching note:** top of the stack, one napkin on the fold spot, start to finish.

Violation: Fold or crease made without a hold

- **Visible cue:** the left gripper is off the cloth while the right gripper folds or creases. Or it lifts away before the crease is finished, and the napkin shifts. A pleat swing is not this violation, because both grippers are on the edge for that.
- **SOP rule broken:** Step 1.3, the crease in Steps 2.3 and 2.4, and Step 2.5. The left gripper presses down and holds while the right gripper works.
- **Coaching note:** hold first, let go last.

Violation: Arms swapped jobs or swung a pleat alone

- **Visible cue:** the left gripper folds on its own or drags a crease. Or the right gripper is used as the hold. Or the wrong arm carries a fan to a plate: the right gripper reaching across to the left plate, or the left gripper going to the middle or right plate. Or one gripper swings the working edge on its own while the other is off the cloth. The edge sags or lands at an angle.
- **SOP rule broken:** Steps 1 and 2, and Steps 2.3 and 2.4. The left gripper holds and the right gripper works, except on the supply lift, the pleat swings, the triangle carry, and the left plate.
- **Coaching note:** left holds, right works. The left plate's fan is the left arm's. Both ends together on a swing and on a triangle.

Violation: Fold made by dragging

- **Visible cue:** a corner or the working edge is pushed, shoved, or slid across the fold spot into place. It is not pinched, lifted off the cloth, and lowered.
- **SOP rule broken:** Steps 1.4, 2.3, 2.4, and 2.5. Lift the cloth off the fold spot before you carry it.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Crease skipped or dabbed

- **Visible cue:** the gripper taps the fold edge in spots, or presses once in the middle, or moves on to the next fold with no stroke at all.
- **SOP rule broken:** Steps 1.5, 2.3, and 2.4, one stroke end to end without lifting off.
- **Coaching note:** one stroke, end to end, no lifting off.

Violation: Pleat swung the wrong way

- **Visible cue:** two pleats in a row go the same way, so the napkin rolls up flat and no zigzag is left.
- **SOP rule broken:** Steps 2.3 and 2.4, every pleat goes the other way from the last one.
- **Coaching note:** away, back, away, back. Say it as you swing.

Violation: Uneven pleats or napkin not pleated to the end

- **Visible cue:** the bands are clearly different widths. Or a crease runs at an angle to the ones next to it, and the episode moves on. Or the gather starts with flat cloth still lying past the last crease. Or one swing is stretched much longer than the rest to use that cloth up.
- **SOP rule broken:** Steps 2.3 and 2.4. Swings the same length, creases the same way as each other, pleats carried to the far edge, and a bad pleat fixed before the gather.
- **Coaching note:** match the last swing before you put the edge down, and keep pleating until there is nothing flat left.

Violation: Gather skipped or done wrong

- **Visible cue:** a fan napkin goes to its plate as a flat pleated band, with the halves never folded onto each other. Or the left gripper holds off the middle, so one half ends up clearly longer. Or the right gripper drags a crease along the gather fold, and the napkin will not open when it leans.
- **SOP rule broken:** Step 2.5. Fold the band in half at the middle so the pleats tuck together, hold the middle and nowhere else, and do not crease this fold.
- **Coaching note:** the pleats are not the fan. The gather is. Middle for the hold, no stroke on that fold.

Violation: Napkin turned or regripped in the air

- **Visible cue:** a napkin turns while a gripper is holding it, between the fold spot and a plate or during a fold. Or a gripper shifts its hold along the cloth without setting the napkin down first.
- **SOP rule broken:** Steps 1.7 and 2.7. Carry it level and lower it as it is. Take a new hold only after the napkin is resting.
- **Coaching note:** put it down before you turn it or take a new hold.

Violation: Fan lifted by the wrong spot

- **Visible cue:** a fan napkin is picked up off the fold spot by a pleat, an open end, or an edge instead of the gather point. The pleats pull apart as it rises.
- **SOP rule broken:** Step 2.6. A fan is pinched at the gather point, never at a pleat or an open end.
- **Coaching note:** fans travel by the middle on one arm. Triangles travel by both ends of the fold edge on two.

Violation: Triangle lifted or carried on one arm

- **Visible cue:** one gripper picks the triangle up off the fold spot on its own, at the middle of the fold edge, at a corner, or at an open side, and carries it to the plate alone. Or both grippers lift it and one lets go on the way, so the triangle swings from a single arm. Or the two grippers pinch the fold edge but one runs ahead of the other, so the triangle tips and a layer slides open.
- **SOP rule broken:** Steps 1.6 and 1.7. A triangle comes up on a U-lift, one gripper on each end of the fold edge, and both arms carry it and let go together.
- **Coaching note:** two ends, one U, all the way to the plate.

Violation: Let go too high or over the wrong place

- **Visible cue:** a gripper opens above the fold spot or above a plate, and the napkin drops the rest of the way. Or a gripper opens over the supply stack, over tabletop away from the fold spot, or over open space.
- **SOP rule broken:** Steps 1 and 2. A gripper opens over the fold spot or a plate and nowhere else, and only once the napkin is resting.
- **Coaching note:** all the way down, and know where it lands before you open.

Violation: Napkin shoved, patted, or pressed down

- **Visible cue:** the gripper nudges, pats, brushes, or straightens a napkin after letting go of it. Or it presses the fan down onto the triangle to make it sit, instead of letting the triangle hold it up.
- **SOP rule broken:** Steps 1 and 2. Fix a bad placement by lifting the napkin clear and laying it down again. Nothing presses a napkin down.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Napkin put on the wrong plate or in the wrong spot

- **Visible cue:** a napkin goes to a plate the turn does not call for. A plate ends up with two triangles, two fans, or three napkins. Or the triangle lands crooked, off center, turned a different way from the other plates, or hanging over the rim. Or the fan is laid flat instead of leaning on the triangle, or is put down before the triangle.
- **SOP rule broken:** Steps 1.7 and 2.7. Plates fill left to right, triangle flat first, fan leaning on it.
- **Coaching note:** name the plate before you lift. Triangle down, fan leaning.

Violation: Napkin dropped, collided, or knocked something over

- **Visible cue:** a napkin falls off the fold spot or a plate. Or a plate or the supply stack is hit and shifts. Or a plate is knocked over. Or a folded napkin falls apart on the way.

- **SOP rule broken:** Steps 1 and 2. Keep each napkin under control on a clear path, and let go only once it is resting.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a napkin still on the fold spot, a plate missing its triangle or its fan, the supply spot not empty, or an arm away from home.
- **SOP rule broken:** Step 3. Check all six napkins on the three plates, the clear fold spot, and the empty supply spot. Return both arms home. Then stop recording.
- **Coaching note:** check first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues. Throw the episode away. Never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, a crash caused by the controller, or a motor error.
- **Bad napkin:** found torn, stained, damp, or stuck to the one below so it cannot be lifted free. Replace it before the next episode.
- **Napkin will not hold a crease:** creased end to end and the fold still pops open every time. Press it before the next episode.
- **Napkin will not hold a fan:** pleated and gathered right and leaned on the triangle right, and the cloth still flops shut because it is too limp to stand up. Use a stiffer napkin.
- **Fan will not stay leaning:** set on the triangle right, and it still slides off every time because the plate is too smooth or too curved. Swap the plate before the next episode.
- **A plate or the supply stack shifts out of its spot** during the episode.

Annotation subtasks (from SOP)

1. Move one napkin from the supply stack to the fold spot
2. Hold a napkin on the fold spot
3. Fold a napkin corner onto the corner across from it
4. Swing one pleat with both arms
5. Crease a fold edge
6. Gather a pleated napkin into a fan
7. U-lift a triangle and lay it flat on a plate with both arms
8. Lean a fan on a triangle

9. Return both arms home and end the episode